TRAIN semantic bank

Nouns / verbs from TRAIN

CLIP text encoder *

Noun bank

Verb bank

Semantic aggregation

Semantic aggregation

01 Regions

patches

CLIP ViT-B/16 Visual encoder *

ROI pooling

Linear projection

V

YOLO detector *

Boxes B



Image I

02 Alignment

Pn

Object alignment

2 stages

Vo

Prompt–region alignment

Q

Masked cross-attention

Linear

Vo′

K,V

CLIP text encoder *

Slot MLPs

Vo

K,V

Q

Masked cross-attention

Linear

Vr′

Pv

Relation alignment

Vr

ROI matching Class + IoU

Image-specific prompts

UPT detector *

Object / relation prompts

UPT boxes

YOLO boxes B

03 Caption

Token prefix

Transformer decoder

Beam search

5 captions

Score candidates

D

C *

O

Weighted selection

Final caption

* Frozen Trainable

D: decoder likelihood C: CLIP alignment O: object consistency